

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO4:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards

(BL2,BL6)

Assessment Tool Weightage: 30%

QUESTIONS: 5

Total Marks:25

Annexure A

Comparative Analysis

Code of Conduct:

I NS BHAVESH(192565986) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

NS BHAVESH(192565986)

Signature of the student:

Annexure A

Comparative Analysis

1. Star topology is generally more expensive than bus topology because it requires a central device such as a switch or hub and more cabling to connect each node individually. In contrast, bus topology uses a single backbone cable, making it low-cost and easier to install. However, in terms of reliability, star topology is superior because failure of one node does not affect others, whereas a failure in the main bus cable can bring down the entire network. Performance is also better in star topology since data transmission is managed through a central device, reducing collisions, while bus topology suffers from performance degradation as more devices are added due to increased data traffic on the shared medium.
2. Mesh topology offers very high fault tolerance because every node is connected to multiple other nodes, providing multiple paths for data transmission. If one link fails, data can still be routed through alternate paths without disrupting communication. In contrast, star topology has moderate fault tolerance; while failure of individual nodes does not affect the network, failure of the central hub or switch can cause the entire network to stop functioning. Therefore, mesh topology is more robust and reliable in critical environments compared to star topology.
3. Bus topology is simple and easy to install because it uses a single backbone cable with minimal connections, making it suitable for small networks. On the other hand, mesh topology is highly complex to install as each node must be connected to every other node, requiring a large number of cables and ports. This complexity increases significantly as the number of devices grows, making mesh topology time-consuming and costly to implement compared to the straightforward setup of bus topology.
4. Hybrid topology is highly scalable because it combines two or more different topologies (such as star, mesh, or bus), allowing the network to expand flexibly according to organizational needs. New segments can be added without affecting the existing network structure. In comparison, star topology is also scalable but to a limited extent, as it depends on the capacity of the central device (switch or hub). Once the central device reaches its port limit, expansion requires additional hardware. Thus, hybrid topology provides greater scalability than star topology.
5. In terms of maintenance, star topology is relatively easy to manage because issues can be quickly identified and isolated to a specific node or cable. Mesh topology, while highly reliable, is difficult to maintain due to its complex structure and large number of connections, making troubleshooting time-consuming. Bus topology is harder to maintain than star because faults in the backbone cable can disrupt the entire network and are difficult to locate. Hybrid topology has moderate maintenance complexity; while it benefits from flexibility, managing multiple combined topologies requires careful monitoring and skilled administration.

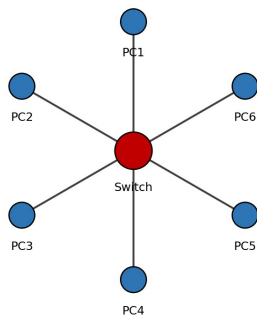
RUBRICS FOR CALCULATION

Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Understanding of Concepts	15%	Demonstrates complete and in-depth understanding of all concepts being compared	Good understanding with minor gaps	Basic understanding; some concepts unclear	Poor understanding of concepts
Comparison Depth	20%	Provides detailed, insightful, and meaningful comparisons across multiple aspects	Adequate comparison with relevant points	Limited comparison; lacks depth	Minimal or incorrect comparison
Criteria Selection	10%	Selects highly relevant and appropriate comparison parameters	Mostly relevant criteria	Some irrelevant or missing criteria	Poor or no criteria selection
Analytical Skills	15%	Strong analysis with logical reasoning and clear justification	Good analysis with some justification	Basic analysis with weak reasoning	No clear analysis
Organization & Structure	10%	Well-organized, clear, and logically structured	Generally organized	Somewhat disorganized	Poorly structured
Use of Examples / Evidence	10%	Uses strong, relevant examples or data to support comparison	Uses some examples	Limited examples	No supporting examples
Clarity of Presentation	10%	Very clear, concise, and easy to understand	Mostly clear	Somewhat unclear	Difficult to understand
Conclusion & Insights	10%	Insightful conclusion with strong justification	Clear conclusion	Basic conclusion	No or weak conclusion

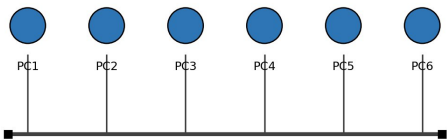
Reference Topology Diagrams

The four base topologies compared throughout this analysis are illustrated below for reference.

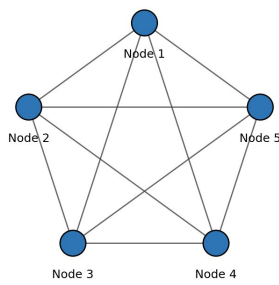
Star Topology



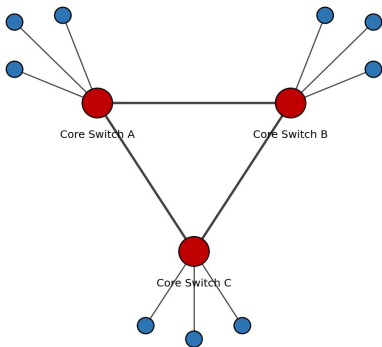
Bus Topology



Mesh Topology (Full Mesh)



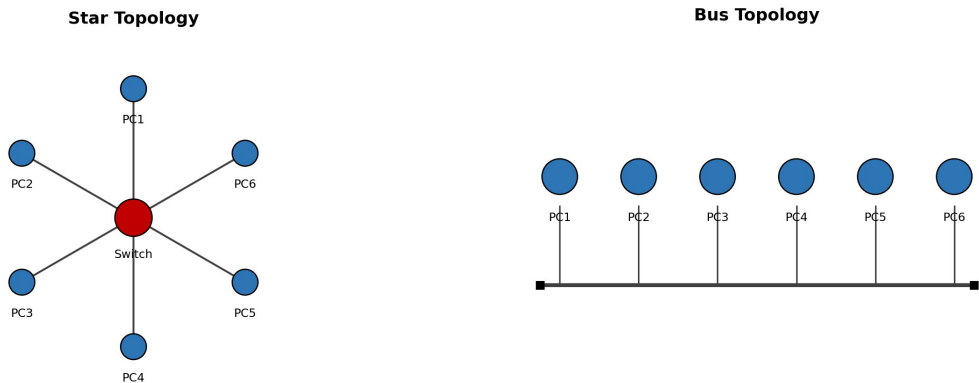
Hybrid Topology (Star-Building Clusters + Mesh Core)



Question 1 — Star vs. Bus Topology (Cost, Reliability, Performance)

Compare Star and Bus topology in terms of cost, reliability, and performance.

Diagrams



Comparative Analysis

Aspect	Star Topology	Bus Topology
Cost	Higher — needs a central switch/hub plus individual cable runs to every node.	Lower — a single shared backbone cable with minimal hardware.
Reliability	Higher — failure of one node's cable/link only isolates that node; other nodes are unaffected.	Lower — a break anywhere in the backbone cable disrupts the entire network.
Performance	Better — the central device manages transmissions, reducing collisions.	Degrades as more devices are added, since all devices share one medium and contend for it, increasing collisions.
Installation	Moderate — more cabling but straightforward per-node wiring.	Simple and quick for small networks.

Analysis / Insight

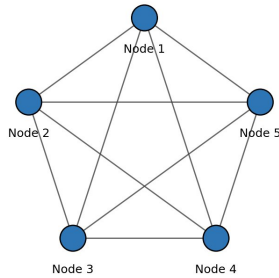
- The cost-vs-reliability trade-off is the central theme: Bus wins on raw hardware cost, but Star wins decisively on fault isolation and consistent performance as the network grows.
- For any network expected to scale beyond a handful of devices, Star's superior reliability and performance typically justify its higher upfront cost.

Question 2 — Mesh vs. Star Topology (Fault Tolerance)

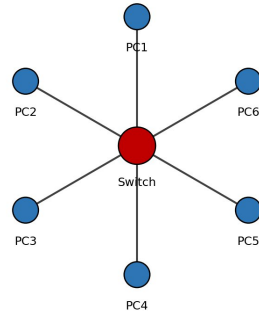
Compare Mesh and Star topology specifically on fault tolerance.

Diagrams

Mesh Topology (Full Mesh)



Star Topology



Comparative Analysis

Aspect	Mesh Topology	Star Topology
Redundant Paths	Every node connects to multiple other nodes, so multiple alternate paths exist between any two points.	Only one path exists between any two nodes — through the central switch/hub.
Link Failure Impact	Data automatically reroutes through an alternate path; communication continues uninterrupted.	A single node-to-switch link failure only isolates that one node — other nodes are unaffected.
Central Device Failure	No single central device exists, so there is no single point of failure at all.	If the central hub/switch fails, the ENTIRE network stops functioning — this is Star's key weakness.
Overall Fault Tolerance	Very high — the most fault-tolerant of the standard topologies.	Moderate — good for individual node failures, but vulnerable at the core.

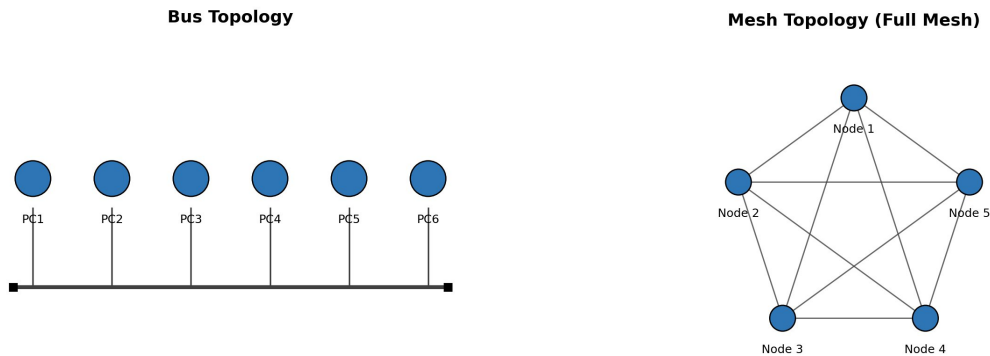
Analysis / Insight

- Mesh is more robust specifically because it has no single point of failure, whereas Star's resilience is entirely dependent on the reliability of one central device.
- This makes Mesh the preferred choice for critical environments (e.g., banking, ISP cores) where even brief central-device downtime is unacceptable.

Question 3 — Bus vs. Mesh Topology (Installation Complexity)

Compare Bus and Mesh topology in terms of installation simplicity and complexity.

Diagrams



Comparative Analysis

Aspect	Bus Topology	Mesh Topology
Cabling Requirement	One single backbone cable with minimal drop connections per node.	Every node must be cabled directly to every other node — cabling grows combinatorially.
Connections for n Nodes	Approximately n connections (one tap per node onto the backbone).	$n(n - 1) / 2$ links for a full mesh — grows very quickly with more devices.
Ports Required	Minimal — often none beyond the shared medium taps.	Each device needs a dedicated port for every other device it connects to.
Installation Time & Cost	Fast and inexpensive — suited to small networks.	Time-consuming and costly, especially as the device count grows.

Analysis / Insight

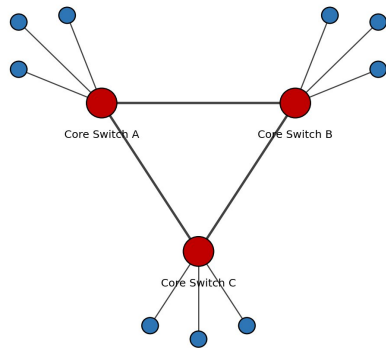
- Bus topology's simplicity comes directly from its shared single-medium design, while Mesh's complexity comes from needing a dedicated physical link between every pair of nodes.
- This is exactly why Mesh is reserved for a small number of core/critical devices (e.g., core routers, switches) rather than being used to connect large numbers of end-user devices.

Question 4 — Hybrid vs. Star Topology (Scalability)

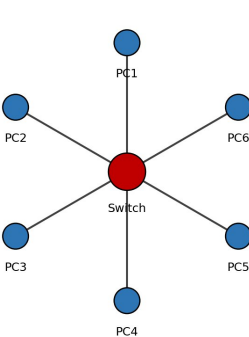
Compare Hybrid and Star topology in terms of scalability for a growing organization.

Diagrams

Hybrid Topology (Star-Building Clusters + Mesh Core)



Star Topology



Comparative Analysis

Aspect	Hybrid Topology	Star Topology
Expansion Method	New segments (new buildings, departments, or topology types) can be added without disrupting the existing structure.	Expansion is limited by the port capacity of the central switch/hub.
Growth Ceiling	Effectively unbounded — additional core/distribution layers can be introduced as needed.	Once the central device's ports are full, new hardware (or a topology redesign) is required.
Flexibility	Combines multiple topology types (star, mesh, bus) so each part of the network can use the design best suited to it.	Limited to the single star pattern throughout.
Best Fit	Large, multi-site, growing organizations (e.g., campuses, enterprises).	Small to medium single-location networks with a known device ceiling.

Analysis / Insight

- Hybrid's scalability advantage comes from its layered structure — growth in one area (e.g., a new building) does not require redesigning the whole network, unlike Star, which hits a hard limit at the central device's port count.
- Hybrid is therefore the standard choice for any organization that expects continued growth across multiple locations or departments.

Question 5 — Maintenance Comparison Across All Four Topologies

Compare Star, Mesh, Bus, and Hybrid topology in terms of maintenance and troubleshooting effort.

Comparative Analysis

Topology	Ease of Maintenance	Why
Star	Easy	Issues can be quickly identified and isolated to a specific node or cable, since each device has its own independent link back to the switch.
Mesh	Difficult	The large number of interconnections makes tracing and troubleshooting a specific fault time-consuming, despite the topology's high reliability.
Bus	Harder than Star	A fault in the shared backbone cable can disrupt the entire network and is often difficult to physically locate along the cable run.
Hybrid	Moderate	Benefits from the flexibility of combined topologies, but managing several different topology types together requires careful monitoring and skilled administration.

Analysis / Insight

- Maintenance difficulty does not track directly with reliability — Mesh is the most fault-tolerant topology but also the hardest to maintain, since redundancy and ease-of-troubleshooting are somewhat opposing design goals.
- Star remains the easiest to maintain precisely because of its centralized, one-link-per-node structure, which is the same property that made it easiest to analyze for fault isolation in Question 1.

Conclusion

- Across all five comparisons, no single topology is universally 'best' — Bus minimizes cost, Star balances manageability and moderate reliability, Mesh maximizes fault tolerance at the cost of complexity, and Hybrid maximizes scalability by combining the strengths of the others where each is most appropriate.
- The correct choice in practice depends on matching the topology's strengths to the specific priority of the deployment: cost (Bus), simplicity/reliability balance (Star), criticality/uptime (Mesh), or large-scale growth (Hybrid).